INCH-POUND
MIL-R-22684/3C
w/ Amendment 1
5 February 2009
SUPERSEDING
MIL-R-22684/3C
6 August 2001

MILITARY SPECIFICATION

# RESISTORS, FIXED, FILM, INSULATED, STYLE RL32

This specification is inactive for design after 2 July 1970. Use MIL-PRF-39017/3.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-22684.

## 1. SCOPE

* 1.1 Scope. This specification covers the associated requirements for insulated, film, fixed resistors of 2-percent and 5-percent resistance tolerance. These resistors are capable of full-load operation at an ambient temperature of 70°C and have a resistance-temperature characteristic of ±200 parts per million per degree Celsius (ppm/°C). Designers are CAUTIONED on using these resistors in high power pulse applications (see 6.3).

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:

M22684/03- 1001

M22684/03- : Performance specification number
1001 : Coded dash number

* The coded dash number is derived in accordance with paragraph 6.2.2 of MIL-PRF-22684.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to Resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.



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2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-22684 - Resistors, Fixed, Film (Insulated), General Specification.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-22684.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Power rating. The power rating shall be 1 watt based on full load operation at an ambient temperature of 70°C.

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 500 volts.

3.5 Resistance values and resistance tolerances. The minimum and maximum standard resistance values and associated resistance tolerances shall be as listed in table I.

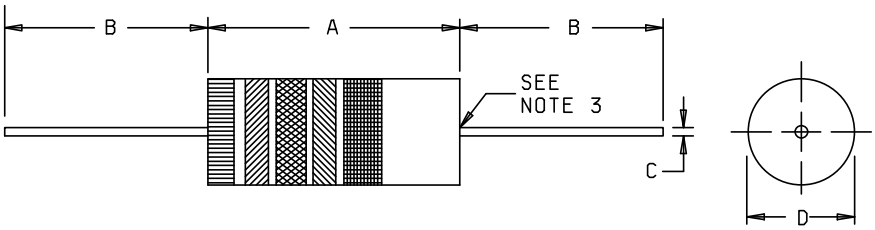
3.6 Terminal type. The terminal type available shall be in accordance with MIL-PRF-22684 and table I.

3.7 Dielectric withstanding voltage. Resistors shall be tested as specified in MIL-PRF-22684. The magnitude of test voltage shall be as follows:

Atmospheric pressure - 1,000 volts rms.
Barometric pressure - 500 volts rms.



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| Ltr | DIMENSIONS | | | |
|---|---|---|---|---|
| | INCHES | | MILLIMETERS | |
| | MIN | MAX | MIN | MAX |
| A | 0.520 | 0.593 | 13.21 | 15.06 |
| B | 1.375 | 1.625 | 34.92 | 41.28 |
| C | .038 | 0.042 | 0.96 | 1.07 |
| D | 0.175 | 0.205 | 4.45 | 5.21 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within .125 inch (3.18 mm) of the resistor body.

FIGURE 1. Configuration and dimensions.

3.8 Insulation resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the insulation resistance shall be not less than 100 megohms.

3.9 Moisture resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 1.5 percent.

3.10 Life. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 2.0 percent.



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Table I. PIN designation.

| Type Designation | Nominal total resistance value | Resistance tolerance (percent) | Terminal | Dash No. | Type Designation | Nominal total resistance value | Resistance tolerance (percent) | Terminal | Dash No. |
|---|---|---|---|---|---|---|---|---|---|
| | Ohms | | | | | Ohms | | | |
| RL32S100G | 10 | G | S | 0001 | RL32S121G | 120 | G | S | 0053 |
| RL32S100J | 10 | J | S | 0002 | RL32S121J | 120 | J | S | 0054 |
| RL32S110G | 11 | G | S | 0003 | RL32S131G | 130 | G | S | 0055 |
| RL32S110J | 11 | J | S | 0004 | RL32S131J | 130 | J | S | 0056 |
| RL32S120G | 12 | G | S | 0005 | RL32S151G | 150 | G | S | 0057 |
| RL32S120J | 12 | J | S | 0006 | RL32S151J | 150 | J | S | 0058 |
| RL32S130G | 13 | G | S | 0007 | RL32S161G | 160 | G | S | 0059 |
| RL32S130J | 13 | J | S | 0008 | RL32S161J | 160 | J | S | 0060 |
| RL32S150G | 15 | G | S | 0009 | RL32S181G | 180 | G | S | 0061 |
| RL32S150J | 15 | J | S | 0010 | RL32S181J | 180 | J | S | 0062 |
| RL32S160G | 16 | G | S | 0011 | RL32S201G | 200 | G | S | 0063 |
| RL32S160J | 16 | J | S | 0012 | RL32S201J | 200 | J | S | 0064 |
| RL32S180G | 18 | G | S | 0013 | RL32S221G | 220 | G | S | 0065 |
| RL32S180J | 18 | J | S | 0014 | RL32S221J | 220 | J | S | 0066 |
| RL32S200G | 20 | G | S | 0015 | RL32S241G | 240 | G | S | 0067 |
| RL32S200J | 20 | J | S | 0016 | RL32S241J | 240 | J | S | 0068 |
| RL32S220G | 22 | G | S | 0017 | RL32S271G | 270 | G | S | 0069 |
| RL32S220J | 22 | J | S | 0018 | RL32S271J | 270 | J | S | 0070 |
| RL32S240G | 24 | G | S | 0019 | RL32S301G | 300 | G | S | 0071 |
| RL32S240J | 24 | J | S | 0020 | RL32S301J | 300 | J | S | 0072 |
| RL32S270G | 27 | G | S | 0021 | RL32S331G | 330 | G | S | 0073 |
| RL32S270J | 27 | J | S | 0022 | RL32S331J | 330 | J | S | 0074 |
| RL32S300G | 30 | G | S | 0023 | RL32S361G | 360 | G | S | 0075 |
| RL32S300J | 30 | J | S | 0024 | RL32S361J | 360 | J | S | 0076 |
| RL32S330G | 33 | G | S | 0025 | RL32S391G | 390 | G | S | 0077 |
| RL32S330J | 33 | J | S | 0026 | RL32S391J | 390 | J | S | 0078 |
| RL32S360G | 36 | G | S | 0027 | RL32S431G | 430 | G | S | 0079 |
| RL32S360J | 36 | J | S | 0028 | RL32S431J | 430 | J | S | 0080 |
| RL32S390G | 39 | G | S | 0029 | RL32S471G | 470 | G | S | 0081 |
| RL32S390J | 39 | J | S | 0030 | RL32S471J | 470 | J | S | 0082 |
| RL32S430G | 43 | G | S | 0031 | RL32S511G | 510 | G | S | 0083 |
| RL32S430J | 43 | J | S | 0032 | RL32S511J | 510 | J | S | 0084 |
| RL32S470G | 47 | G | S | 0033 | RL32S561G | 560 | G | S | 0085 |
| RL32S470J | 47 | J | S | 0034 | RL32S561J | 560 | J | S | 0086 |
| RL32S510G | 51 | G | S | 0035 | RL32S621G | 620 | G | S | 0087 |
| RL32S510J | 51 | J | S | 0036 | RL32S621J | 620 | J | S | 0088 |
| RL32S560G | 56 | G | S | 0037 | RL32S681G | 680 | G | S | 0089 |
| RL32S560J | 56 | J | S | 0038 | RL32S681J | 680 | J | S | 0090 |
| RL32S620G | 62 | G | S | 0039 | RL32S751G | 750 | G | S | 0091 |
| RL32S620J | 62 | J | S | 0040 | RL32S751J | 750 | J | S | 0092 |
| RL32S680G | 68 | G | S | 0041 | RL32S821G | 820 | G | S | 0093 |
| RL32S680J | 68 | J | S | 0042 | RL32S821J | 820 | J | S | 0094 |
| RL32S750G | 75 | G | S | 0043 | RL32S911G | 910 | G | S | 0095 |
| RL32S750J | 75 | J | S | 0044 | RL32S911J | 910 | J | S | 0096 |
| RL32S820G | 82 | G | S | 0045 | RL32S102G | 1,000 | G | S | 0097 |
| RL32S820J | 82 | J | S | 0046 | RL32S102J | 1,000 | J | S | 0098 |
| RL32S910G | 91 | G | S | 0047 | RL32S112G | 1,100 | G | S | 0099 |
| RL32S910J | 91 | J | S | 0048 | RL32S112J | 1,100 | J | S | 0100 |
| RL32S101G | 100 | G | S | 0049 | RL32S122G | 1,200 | G | S | 0101 |
| RL32S101J | 100 | J | S | 0050 | RL32S122J | 1,200 | J | S | 0102 |
| RL32S111G | 110 | G | S | 0051 | RL32S132G | 1,300 | G | S | 0103 |
| RL32S111J | 110 | J | S | 0052 | RL32S132J | 1,300 | J | S | 0104 |



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Table I. PIN designation - continued.

| Type Designation | Nominal total resistance value | Resistance tolerance (percent) | Terminal | Dash No. | Type Designation | Nominal total resistance value | Resistance tolerance (percent) | Terminal | Dash No. |
|---|---|---|---|---|---|---|---|---|---|
| | Ohms | | | | | Ohms | | | 0159 |
| RL32S152G | 1,500 | G | S | 0105 | RL32S203G | 20,000 | G | S | 0160 |
| RL32S152J | 1,500 | J | S | 0106 | RL32S203J | 20,000 | J | S | 0161 |
| RL32S162G | 1,600 | G | S | 0107 | RL32S223G | 22,000 | G | S | 0162 |
| RL32S162J | 1,600 | J | S | 0108 | RL32S223J | 22,000 | J | S | |
| RL32S182G | 1,800 | G | S | 0109 | RL32S243G | 24,000 | G | S | 0163 |
| RL32S182J | 1,800 | J | S | 0110 | RL32S243J | 24,000 | J | S | 0164 |
| RL32S202G | 2,000 | G | S | 0111 | RL32S273G | 27,000 | G | S | 0165 |
| RL32S202J | 2,000 | J | S | 0112 | RL32S273J | 27,000 | J | S | 0166 |
| RL32S222G | 2,200 | G | S | 0113 | RL32S303G | 30,000 | G | S | 0167 |
| RL32S222J | 2,200 | J | S | 0114 | RL32S303J | 30,000 | J | S | 0168 |
| RL32S242G | 2,400 | G | S | 0115 | RL32S333G | 33,000 | G | S | 0169 |
| RL32S242J | 2,400 | J | S | 0116 | RL32S333J | 33,000 | J | S | 0170 |
| RL32S272G | 2,700 | G | S | 0117 | RL32S363G | 36,000 | G | S | 0171 |
| RL32S272J | 2,700 | J | S | 0118 | RL32S363J | 36,000 | J | S | 0172 |
| RL32S302G | 3,000 | G | S | 0119 | RL32S393G | 39,000 | G | S | 0173 |
| RL32S302J | 3,000 | J | S | 0120 | RL32S393J | 39,000 | J | S | 0174 |
| RL32S332G | 3,300 | G | S | 0121 | RL32S433G | 43,000 | G | S | 0175 |
| RL32S332J | 3,300 | J | S | 0122 | RL32S433J | 43,000 | J | S | 0176 |
| RL32S362G | 3,600 | G | S | 0123 | RL32S473G | 47,000 | G | S | 0177 |
| RL32S362J | 3,600 | J | S | 0124 | RL32S473J | 47,000 | J | S | 0178 |
| RL32S392G | 3,900 | G | S | 0125 | RL32S513G | 51,000 | G | S | 0179 |
| RL32S392J | 3,900 | J | S | 0126 | RL32S513J | 51,000 | J | S | 0180 |
| RL32S432G | 4,300 | G | S | 0127 | RL32S563G | 56,000 | G | S | 0181 |
| RL32S432J | 4,300 | J | S | 0128 | RL32S563J | 56,000 | J | S | 0182 |
| RL32S472G | 4,700 | G | S | 0129 | RL32S623G | 62,000 | G | S | 0183 |
| RL32S472J | 4,700 | J | S | 0130 | RL32S623J | 62,000 | J | S | 0184 |
| RL32S512G | 5,100 | G | S | 0131 | RL32S683G | 68,000 | G | S | 0185 |
| RL32S512J | 5,100 | J | S | 0132 | RL32S683J | 68,000 | J | S | 0186 |
| RL32S562G | 5,600 | G | S | 0133 | RL32S753G | 75,000 | G | S | 0187 |
| RL32S562J | 5,600 | J | S | 0134 | RL32S753J | 75,000 | J | S | 0188 |
| RL32S622G | 6,200 | G | S | 0135 | RL32S823G | 82,000 | G | S | 0189 |
| RL32S622J | 6,200 | J | S | 0136 | RL32S823J | 82,000 | J | S | 0190 |
| RL32S682G | 6,800 | G | S | 0137 | RL32S913G | 91,000 | G | S | 0191 |
| RL32S682J | 6,800 | J | S | 0138 | RL32S913J | 91,000 | J | S | 0192 |
| RL32S752G | 7,500 | G | S | 0139 | | | | | |
| RL32S752J | 7,500 | J | S | 0140 | | Megohm | | | |
| RL32S822G | 8,200 | G | S | 0141 | RL32S104G | 0.10 | G | S | 0193 |
| RL32S822J | 8,200 | J | S | 0142 | RL32S104J | 0.10 | J | S | 0194 |
| RL32S912G | 9,100 | G | S | 0143 | RL32S114G | 0.11 | G | S | 0195 |
| RL32S912J | 9,100 | J | S | 0144 | RL32S114J | 0.11 | J | S | 0196 |
| RL32S103G | 10,000 | G | S | 0145 | RL32S124G | 0.12 | G | S | 0197 |
| RL32S103J | 10,000 | J | S | 0146 | RL32S124J | 0.12 | J | S | 0198 |
| RL32S113G | 11,000 | G | S | 0147 | RL32S134G | 0.13 | G | S | 0199 |
| RL32S113J | 11,000 | J | S | 0148 | RL32S134J | 0.13 | J | S | 0200 |
| RL32S123G | 12,000 | G | S | 0149 | RL32S154G | 0.15 | G | S | 0201 |
| RL32S123J | 12,000 | J | S | 0150 | RL32S154J | 0.15 | J | S | 0202 |
| RL32S133G | 13,000 | G | S | 0151 | RL32S164G | 0.16 | G | S | 0203 |
| RL32S133J | 13,000 | J | S | 0152 | RL32S164J | 0.16 | J | S | 0204 |
| RL32S153G | 15,000 | G | S | 0153 | RL32S184G | 0.18 | G | S | 0205 |
| RL32S153J | 15,000 | J | S | 0154 | RL32S184J | 0.18 | J | S | 0206 |
| RL32S163G | 16,000 | G | S | 0155 | RL32S204G | 0.20 | G | S | 0207 |
| RL32S163J | 16,000 | J | S | 0156 | RL32S204J | 0.20 | J | S | 0208 |
| RL32S183G | 18,000 | G | S | 0157 | RL32S224G | 0.22 | G | S | 0209 |
| RL32S183J | 18,000 | J | S | 0158 | RL32S224J | 0.22 | J | S | 0210 |



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Table I. PIN designation - continued.

| Type Designation | Nominal total resistance value | Resistance tolerance (percent) | Terminal | Dash No. | Type Designation | Nominal total resistance value | Resistance tolerance (percent) | Terminal | Dash No. |
|---|---|---|---|---|---|---|---|---|---|
| | Megohm | | | | | Megohm | | | |
| RL32S244G | 0.24 | G | S | 0211 | RL32S514G | 0.51 | G | S | 0227 |
| RL32S244J | 0.24 | J | S | 0212 | RL32S514J | 0.51 | J | S | 0228 |
| RL32S274G | 0.27 | G | S | 0213 | RL32S564G | 0.56 | G | S | 0229 |
| RL32S274J | 0.27 | J | S | 0214 | RL32S564J | 0.56 | J | S | 0230 |
| RL32S304G | 0.30 | G | S | 0215 | RL32S624G | 0.62 | G | S | 0231 |
| RL32S304J | 0.30 | J | S | 0216 | RL32S624J | 0.62 | J | S | 0232 |
| RL32S334G | 0.33 | G | S | 0217 | RL32S684G | 0.68 | G | S | 0233 |
| RL32S334J | 0.33 | J | S | 0218 | RL32S684J | 0.68 | J | S | 0234 |
| RL32S364G | 0.36 | G | S | 0219 | RL32S754G | 0.75 | G | S | 0235 |
| RL32S364J | 0.36 | J | S | 0220 | RL32S754J | 0.75 | J | S | 0236 |
| RL32S394G | 0.39 | G | S | 0221 | RL32S824G | 0.82 | G | S | 0237 |
| RL32S394J | 0.39 | J | S | 0222 | RL32S824J | 0.82 | J | S | 0238 |
| RL32S434G | 0.43 | G | S | 0223 | RL32S914G | 0.91 | G | S | 0239 |
| RL32S434J | 0.43 | J | S | 0224 | RL32S914J | 0.91 | J | S | 0240 |
| RL32S474G | 0.47 | G | S | 0225 | RL32S105G | 1.0 | G | S | 0241 |
| RL32S474J | 0.47 | J | S | 0226 | RL32S105J | 1.0 | J | S | 0242 |

## 4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with Group A inspection and Group B inspection of MIL-PRF-22684 and as specified herein.

## 5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes The notes specified in MIL-PRF-22684 are applicable to this specification.



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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, and complete PIN (see 1.2).

* b. If not otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.

c. Packaging requirements.

* 6.3 Pulse applications. Designers are CAUTIONED on using these resistors in high power pulse applications. Since they have not been qualified nor tested for such applications, damage and premature failure are possible. These resistors only see a one time pulse (Short-time overload) as part of the group B inspection of this specification. Designers MAY CONSIDER using DSCC drawings 99001, and 03006 for high power pulse applications.

* 6.4 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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